

Notes on Variance Calculations

The mathematical derivations showing how we can set achieve the following in our simulations:

- The variance of X , Y and C is 1.
- Each variable has a mean 0.
- The autoregressive effects of X_{t-1} on X_t is 0.25 (same for Y_{t-1} on Y_t).
- The lag-1 cross effect of X_{t-1} on Y_t is 0.10 (same for Y_{t-1} on X_t).
- There are 3 confounders, and hence $k = 3$.
- Their effects differ for X and Y , such that X is affected at $[0.30, 0.20, 0.10]$ and Y is affected at $[0.30, 0.20, 0.10]$.
- The innovations have residual correlations of 0.3 at each time-point, but not across time-points.
- At the first time-point ($T = 0$), X and Y are correlated due to the confounding variables.
- At the first time-point, we can assume that the dynamic process was going on before the data started being collected. Hence we want to simulate a system that is already stationary, but also where the first time-point has some 'extra' correlation of X and Y due to the fact that we assume the process was going before the start of the data collection.

Defining matrices

We have variables X over occasions T and Y over occasions T . This yields the vector:

$$Z_t = \begin{bmatrix} X_t \\ Y_t \end{bmatrix}$$

For one individual i , the vector looks like:

$$Z_i = \begin{bmatrix} X_{i1} \\ Y_{i1} \\ X_{i2} \\ Y_{i2} \\ X_{i3} \\ Y_{i3} \\ X_{i4} \\ Y_{i4} \\ X_{i5} \\ Y_{i5} \end{bmatrix}.$$

Expanding for N individuals:

$$Z = \begin{bmatrix} X_{1,1} & Y_{1,1} & X_{1,2} & Y_{1,2} & X_{1,3} & Y_{1,3} & X_{1,4} & Y_{1,4} & X_{1,5} & Y_{1,5} \\ X_{2,1} & Y_{2,1} & X_{2,2} & Y_{2,2} & X_{2,3} & Y_{2,3} & X_{2,4} & Y_{2,4} & X_{2,5} & Y_{2,5} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ X_{N,1} & Y_{N,1} & X_{N,2} & Y_{N,2} & X_{N,3} & Y_{N,3} & X_{N,4} & Y_{N,4} & X_{N,5} & Y_{N,5} \end{bmatrix}.$$

Now for time-point t , X_t and Y_t are made up of some part of the confounder effect, and some part by the dynamic effects. With the dynamic effects here we mean every effect that 'feeds forward' to the next time-point. For the first time-point, there are no dynamic effects at all, since there are no previous time-points. Another way of phrasing this is that some part of the variance in X (or Y) is explained by the confounders, some part is explained by the dynamic effects, and some part is explained by the innovations. We can capture this in the formula:

$$Z_t = AZ_{t-1} + BU + E_t,$$

where

- A is the matrix of autoregressive and cross-lagged effects,
- B is the matrix of confounder effects,
- U is the vector of time-invariant confounders,
- E_t is the vector of innovations at time-point t .

If we choose our effects as denoted above we get the following A matrix:

$$A = \begin{bmatrix} a_x & b_y \\ b_x & a_y \end{bmatrix} = \begin{bmatrix} 0.25 & 0.10 \\ 0.10 & 0.25 \end{bmatrix}$$

We get the following beta matrix:

$$B = \begin{bmatrix} c_{x1} & c_{x2} & c_{x3} \\ c_{y1} & c_{y2} & c_{y3} \end{bmatrix} = \begin{bmatrix} 0.30 & 0.20 & 0.10 \\ 0.30 & 0.20 & 0.10 \end{bmatrix}$$

And matrix U :

$$U = \begin{bmatrix} c_{1,1} & c_{1,2} & c_{1,3} \\ c_{2,1} & c_{2,2} & c_{2,3} \\ c_{3,1} & c_{3,2} & c_{3,3} \\ \vdots & \vdots & \vdots \\ c_{N,1} & c_{N,2} & c_{N,3} \end{bmatrix}$$

Finally, we can specify the innovations matrix as:

$$E_t = \begin{bmatrix} e_{X_t} \\ e_{Y_t} \end{bmatrix}$$

Calculating variances part 1

As we just denoted, each measurement occasion takes some variance from the confounders and some from the dynamic effect. The variance received from the confounders is time-invariant, which means we can calculate it directly. Since the variances of all variables (and thus also the confounders) is 1, and the confounders are not correlated with each other, their covariance matrix is the identity matrix I . Hence, the variance received from the confounders is:

$$S_U = B\Psi B^T = BIB^T = BB^T$$

If we use the values from above, we get:

$$S_U = \begin{bmatrix} 0.30 & 0.20 & 0.10 \\ 0.30 & 0.20 & 0.10 \end{bmatrix} \begin{bmatrix} 0.30 & 0.30 \\ 0.20 & 0.20 \\ 0.10 & 0.10 \end{bmatrix} = \begin{bmatrix} 0.14 & 0.14 \\ 0.14 & 0.14 \end{bmatrix}$$

The confounders add a variance of 0.14 to both X and Y, and a covariance of 0.14 between X and Y at each time-point. We wanted each variable to have a variance of 1, which means that the dynamic effect still needs to add $1 - 0.14 = 0.86$ to the variance of each variable. We also want the innovations to have a correlation of 0.3 at each time-point, which means that the part of X and Y that cannot be explained by $X_t - 1$ and $Y_t - 1$ or C needs to be correlated at 0.3. In math notation, we want:

$$\text{Corr}(e_{X_t}, e_{Y_t}) = \rho = 0.3$$

We can denote the variance of the innovations as:

$$\text{Var}(e_{X_t}) = v_1$$

and

$$\text{Var}(e_{Y_t}) = v_2$$

The covariance between the innovations is then their correlation multiplied by their standard deviations:

$$\text{Cov}(e_{X_t}, e_{Y_t}) = \rho\sqrt{v_1}\sqrt{v_2}$$

The covariance matrix of the innovations can then be written as:

$$S_E = \begin{bmatrix} v_1 & \rho\sqrt{v_1}\sqrt{v_2} \\ \rho\sqrt{v_1}\sqrt{v_2} & v_2 \end{bmatrix}$$

Since we know that rho is 0.3, we can rewrite this as:

$$S_E = \begin{bmatrix} v_1 & 0.3\sqrt{v_1}\sqrt{v_2} \\ 0.3\sqrt{v_1}\sqrt{v_2} & v_2 \end{bmatrix}$$

Which leaves us with two unknown values: v_1 and v_2 . We need to choose these such that the total variance of X and Y becomes 1. Another way of phrasing this requirement is that S , which is the total variance-covariance matrix of Z_t , needs to have 1s on the diagonal.

Calculating variances part 2

Now first we need to specify what S looks like. We know that:

$$S = \text{var}(Z_t) = \begin{bmatrix} \text{var}(X_t) & \text{cov}(X_t, Y_t) \\ \text{cov}(X_t, Y_t) & \text{var}(Y_t) \end{bmatrix}$$

We want the variance of X and Y to be 1, which means that we want:

$$\text{var}(X_t) = \text{var}(Y_t) = 1$$

, yielding:

$$S = \begin{bmatrix} 1 & \text{cov}(X_t, Y_t) \\ \text{cov}(X_t, Y_t) & 1 \end{bmatrix}$$

We can now denote $\text{cov}(X_t, Y_t)$ as c . This yields:

$$S = \begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix}$$

Note that this covariance matrix is stationary when it is the same for all time-points. In order to satisfy this constraint, the following equation must hold:

$$S = ASA^T + S_U + S_E$$

The covariance matrix of the current time-point, is equal to that of the previous time-point, multiplied by the cross-lagged and autoregressive effects, plus the variance received from the confounders, plus the variance received from the innovations. We must solve this equation for v_1 , v_2 and c . v_1 and v_2 are only present in S_E , while c is only present in S . Expanding the equation yields:

$$\begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix} = \begin{bmatrix} (ASA^T)_{11} & (ASA^T)_{12} \\ (ASA^T)_{21} & (ASA^T)_{22} \end{bmatrix} + \begin{bmatrix} 0.14 & 0.14 \\ 0.14 & 0.14 \end{bmatrix} + \begin{bmatrix} v_1 & \rho\sqrt{v_1v_2} \\ \rho\sqrt{v_1v_2} & v_2 \end{bmatrix}.$$